

M. TECH
(SEM I) THEORY EXAMINATION 2022-23
DATA WAREHOUSING AND DATA MINING

Time: 3 Hours**Total Marks: 70****Note:** Attempt all Sections. If require any missing data; then choose suitably.**SECTION A****1. Attempt all questions in brief. 2 x 7 = 14**

- (a) Define Data Warehouse.
- (b) Explain Data Cube.
- (c) Discuss difference between OLTP and OLAP.
- (d) Discuss Entropy with example.
- (e) Discuss basic assumption in Naïve Bayes Classifiers.
- (f) Discuss difference between clustering and classification.
- (g) Explain Spatial Data with example.

SECTION B**2. Attempt any three of the following: 7 x 3 = 21**

- (a) Discuss different types of schemas in Data warehousing.
- (b) Explain different type of Normalization with suitable example.
- (c) Design Decision Tree using ID3 algorithm by taking a suitable example.
- (d) Discuss K means algorithm with proper example.
- (e) Explain descriptive mining of complex data objects.

SECTION C**3. Attempt any one part of the following: 7 x 1 = 7**

- (a) Describe Three-tier data warehouse architecture with a neat diagram.
- (b) Describe different types of OLAP operations with examples.

4. Attempt any one part of the following: 7 x 1 = 7

- (a) Explain Data Transformation with suitable example.

(b)	Transactions	List of Items
	T1	I1, I2, I3, I5
	T2	I2, I3, I4
	T3	I4, I5
	T4	I1, I2, I4
	T5	I1, I2, I3, I5
	T6	I1, I2, I3, I4

Support=50% Confidence=60%

Generate association rule for given items using Apriori Algorithms

5. Attempt any *one* part of the following: **7 x 1 = 7**

- (a) Illustrate Naïve Bayes classifier with suitable example
- (b) Design a classifier using SVM by taking a proper dataset.

6. Attempt any *one* part of the following: **7 x 1 = 7**

- (a) Discuss DBSCAN algorithm for clustering.
- (b) Illustrate Hierarchical clustering algorithm with examples.

7. Attempt any *one* part of the following: **7 x 1 = 7**

- (a) Describe web mining and its types with proper example.
- (b) Describe multimedia data mining and its types with proper example.

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